Tristan Metz



Gameplay & Tools Programmer

Lead programmer at XR Japan in Tokyo. I build the systems, editors and frameworks that let a team move fast, from a custom C++ engine to a medical simulator used to train doctors.

Basic info

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LinkedIn
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GitHub
github.com/123tris

itch.io
123tris.itch.io

Location
Tokyo, Japan

Citizenship
Dutch (EU)

Education
Bachelor of Creative Media and Game Technologies
Saxion University of Applied Sciences, Enschede, the Netherlands

Skills

Gameplay & systems
Character controllers, game feel, VR interaction, local multiplayer, dialogue systems, UI, audio integration

Tools & architecture
Unity editor tooling, ScriptableObject architecture, shared Unity packages, level editors, serialization & reflection, debugging tools

Engines
Unity (since 2016), custom C++ engines (Vulkan, OpenGL), Unreal Engine

Programming languages
C#, C++, HLSL/ShaderLab, Lua

Frameworks & libraries
URP, Input System, XR Interaction Toolkit, Mirror, Burst/Jobs, Odin Inspector, DOTween, FMOD, ImGui, Vulkan

Tools

Summary

I am a game programmer specializing in gameplay systems and developer tooling, working mostly in C# and Unity, with a background in C++ engine development. I like building the frameworks and editors that make the rest of a team faster, and then shipping games with them.

At XR Japan in Tokyo I lead programming on Lumen, a virtual endoscopy trainer in which doctors practise endoscopic surgery with a custom controller, as well as VR and iPadOS applications for other clients: architecture, simulation and performance work, shared libraries and tools, technical discussions with clients and mentoring the team. Outside work I co-founded Idomi Games, where I program Ryze, a precision platformer on Steam that began as a Global Game Jam entry, and I maintain Ripple, an open-source ScriptableObject framework.

Work history

Lead Programmer

March 2025 — Present
XR Japan (formerly AVR Japan), Tokyo, Japan

- Lead programmer on Lumen, a virtual endoscopy trainer for practising endoscopic submucosal dissection (ESD), and on VR and iPadOS training applications for clients
- Tissue simulation, lesion marking and the input from Lumen's custom endoscope controller; Burst-compiled solvers and faster mesh-data loading
- Own the technical side of client projects: technical discussions with clients, estimates, progress and risk tracking
- Mentor and onboard programmers; set the architecture and conventions projects are built on
- Built the shared Unity packages used across projects: Ripple (events and state), a VR template, UnityUtilities and AutoAssign
- Set up test environments and tooling that remove repetitive work from the team

Game Programmer

September 2024 — February 2025
AVR Japan, Tokyo, Japan

- Gameplay, UI and tools for VR and tablet training applications in Unity, with some Unreal Engine work
- Created the company's VR template package: reusable interaction, fade, tween and conditional-event components

Co-founder & Programmer

January 2022 — Present
Idomi Games (indie studio, alongside full-time work)

- Programmer of Ryze, a 2D precision platformer with a Steam playtest out now
- The character controller and throw-to-teleport movement, the game's core feel
- Level mechanics, from moving and crumbling platforms to pistons and conveyor belts, and a test level that exercises every one of them
- A reusable dialogue system with NPCs, cutscenes and localization
- Programmer on the studio's game jam entries, including Shroomer (GMTK Game Jam 2026) and This Side Up (Mini Jam 188)

Git, Azure DevOps, Rider, Visual Studio, CMake

Languages
English (fluent), Dutch (native), Japanese (JLPT N1)

Gameplay Programmer (graduation internship)
September 2021 — April 2022
VR Amsterdam, Amsterdam, the Netherlands

- Sole programmer of Ghostebash, a free-roaming multiplayer VR arcade game, starting from an empty project
- Real-time local multiplayer with Mirror; IK rigs that map players' real bodies into the game; physics-based interactions with the XR Interaction Toolkit
- Graduation research into hardware, networking and player retention for free-roaming VR, including measuring network cost per concurrent player

Programmer Intern
February 2019 — July 2019
Damocles Games, the Netherlands

- Turn-based strategy game on a 3D grid (Fire Emblem-like) in Unity, in a 10-person team
- Built an API and custom visual-scripting units so designers could create gameplay behaviours themselves, speeding up playtest iteration
- Rewrote the selection and highlight framework, which made "simple" features like pawn hovering and turn-based camera controls quick to add
- Reduced the old codebase's size while making it more flexible; worked directly with the UX and game designers

Selected projects

- Lumen — virtual endoscopy trainer, XR Japan (in beta)
- Ryze — 2D precision platformer, Unity 6 (2022 — present)
- Ripple — open-source ScriptableObject framework for Unity
- BotBandJazz — VR game shown at the Dutch Game Garden (2017)
- Tristeon — C++ game engine with a Vulkan renderer and editor (2017)
- osu!catch level editor — community tool, Unity (2018 — 2019)